

Find every prime up to 100

Cross out the numbers that can be shared into equal groups. Whatever is still standing at the end is a prime number.

1. Cross out 1. It is not a prime, because a prime needs exactly two divisors and 1 has only itself.
2. Circle 2. Then colour every other multiple of 2: 4, 6, 8, 10 and so on.
3. Circle 3, the first number left. Colour every multiple of 3: 6, 9, 12, 15 ...
4. Circle 5 and colour its multiples, then circle 7 and colour its multiples.
5. Stop. Every number left uncoloured is prime. Count them: there should be 25.

Colour key:  multiples of 2  multiples of 3  multiples of 5  multiples of 7

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

Why stop at 7? The next prime is 11, and $11 \times 11 = 121$, which is past 100 already. Every composite number under 100 has a factor of 2, 3, 5 or 7, so they are all gone by now.

Prime or not?

For each number, try to circle the dots into equal groups with none left over. If you can, the number is not prime. Write the two factors you found.

6 ○○○○○○

☐ prime ☐ not prime because _____

9 ○○○○○○○○

☐ prime ☐ not prime because _____

11 ○○○○○○○○○○

☐ prime ☐ not prime because _____

12 ○○○○○○○○○○

☐ prime ☐ not prime because _____

15 ○○○○○○○○○○○○

☐ prime ☐ not prime because _____

17 ○○○○○○○○○○○○

☐ prime ☐ not prime because _____

18 ○○○○○○○○○○○○

☐ prime ☐ not prime because _____

21 ○○○○○○○○○○○○

☐ prime ☐ not prime because _____

23 ○○○○○○○○○○○○

☐ prime ☐ not prime because _____

Your turn

Pick three numbers of your own, draw that many dots, and test them the same way.

☐ prime ☐ not prime because _____☐ prime ☐ not prime because _____☐ prime ☐ not prime because _____

How to test any number under 100. Try dividing by 2, then 3, then 5, then 7. If none of them divide it evenly, the number is prime. You never need to try anything bigger, because two factors of 11 or more would already multiply past 100.